IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-019

1. Topic (12 words max)

AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Coconut is a strategically important plantation crop in Sri Lanka, and the Coconut Research Institute of Sri Lanka (CRISL) has a key role in conserving coconut genetic resources and addressing disease threats. However, present coconut GenBank management and disease surveillance processes are primarily manual, fragmented, and minimally supported by intelligent digital tools. CRISL GenBank lack a centralized geographical framework for correctly mapping coconut tree distribution, evaluating land use, and identifying regionally specific unoccupied sites for plantation growth. Existing land evaluations are unstructured, time consuming, and do not enable data driven, long term planning decisions.

In parallel, coconut germplasm identification and record management are mainly dependent on limits scalability and consistency among national gene banks. Field photos are not systematically analyzed to extract CRI defined morphological criteria, and there is no standardized AI-powered framework for converting visual data into structured, verifiable digital genebank records. Furthermore, traceability, access control, and audit systems for image-based germplasm data are insufficient.

Coconut disease surveillance raises similar issues. Monitoring systems are generally expert driven and periodic, leading in late diagnosis of new illness problems. Despite of the huge amount of crowdsourced photographs and reports available, there is no uniform platform for integrating real time observations from farmers, field officers, students, and academics. This unstructured data is still unused for actionable intelligence.

Weligama Coconut Leaf Wilt Disease (WCLWD) is a particularly serious threat to coconut productivity. Existing IT-based techniques mostly focus on disease classification rather than severity evaluation, which means they struggle to differentiate WCLWD from visually comparable nutrient deficiencies. Advanced segmentation methods require pixel-level annotation, which is costly and not practical for large scale deployment, leaving a key gap in scalable severity measurement.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products and Systems

Existing agricultural information systems are mostly standalone solutions that address only individual problems such as disease detection, land surveying, or germplasm documentation. Most commercial and institutional applications rely on manual field observations, non-georeferenced data, and expert-based visual assessments. In Sri Lanka, coconut gene bank land utilization, germplasm identification, and disease monitoring are carried out through small-scale, partially manual workflows with limited automation, minimal GIS integration, and no unified intelligence platform. These systems lack scalability, real-time analytical capability, and secure end-to-end digital traceability.

Existing Research

Deep learning models such as CNNs, Vision Transformers (ViT), and explainable AI techniques like Grad-CAM have shown strong performance in plant disease detection. However, most research focuses on image-level classification rather than spatial or severity-aware analysis. Segmentation-based severity estimation is limited by the need for costly and time-consuming pixel-level annotations. Furthermore, GIS-driven satellite/drone analytics, crowdsourced disease surveillance, explainable AI, and AI-based germplasm digitization are typically investigated in isolation, with limited integration of spatial intelligence and institutional data governance.

Gaps/Limitations Identified

Despite these advances, significant gaps remain across all four proposed components. Existing systems lack a satellite / drone-based, GIS-centric land utilization framework capable of accurately mapping coconut gene banks, identifying spatially precise vacant land, and supporting long-term plantation planning. There is also no standardized, secure, AI-driven system to convert coconut palm and nut images into verifiable and auditable digital germplasm records. Current disease monitoring approaches lack real-time community intelligence, leading to delayed detection of emerging threats. Additionally, disease severity assessment methods rely on computationally heavy models and manual pixel-level annotation, limiting scalability. The proposed research addresses these gaps through integrated GIS analytics, explainable weakly supervised learning, secure digital workflows, and community-driven agricultural intelligence tailored to Sri Lankan coconut research needs.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

In order to assist the Coconut Research Institute of Sri Lanka (CRISL) with coconut gene bank management, disease surveillance, and conservation decision-making, this study proposes an integrated, AI-driven digital intelligence framework. The solution combines GIS, satellite/drone technology, computer vision, machine learning, and community intelligence platforms to overcome key limitations identified in Section 6, including manual procedures, poor spatial accuracy, limited scalability, delayed disease detection, and the absence of centralized, structured digital records.

The proposed solution addresses these gaps through four tightly coupled components.

GIS-Driven Land Utilization Analysis for Coconut Gene Banks employs satellite/drone imagery and GIS layers to accurately map coconut tree distribution, identify vacant land, analyze density patterns, and calculate precise plantable areas. This eliminates non-georeferenced, manual land assessments and supports evidence-based land-use planning and long-term genebank expansion decisions.

Secure AI-Based Germplasm Identification System replaces manual visual identification with a centralized, CRI-aligned image classification platform. CNN-based models are automatically applied to coconut palm and nut images to extract standardized morphological parameters, assign unique Palm IDs, and store verified digital records with role-based access control and audit logging.

Crowdsourcing-Based Community Intelligence Platform overcomes delayed and expert-only disease monitoring by enabling farmers, field officers, students, and researchers to contribute real-time field observations. Low-data AI techniques transform unstructured images and reports into structured intelligence, supporting early disease outbreak detection and continuous gene bank condition monitoring.

WCLWD Severity Assessment Framework integrates weakly supervised semantic segmentation with the ABCD severity assessment rule to address the lack of accurate severity quantification. This approach enables pixel-level identification of diseased regions without expensive manual annotation and improves differentiation between WCLWD symptoms and visually similar nutrient deficiencies.

Together, these components form a scalable and unified digital ecosystem rather than isolated tools. Following are the inputs, process and outputs of our research:

Inputs

- Satellite-captured / Drone-captured aerial images from genebank sites
- Field-captured images of coconut palms, nuts, and leaves
- Crowdsourced community reports, images, and textual observations
- Expert-verified CRI reference data and disease guidelines

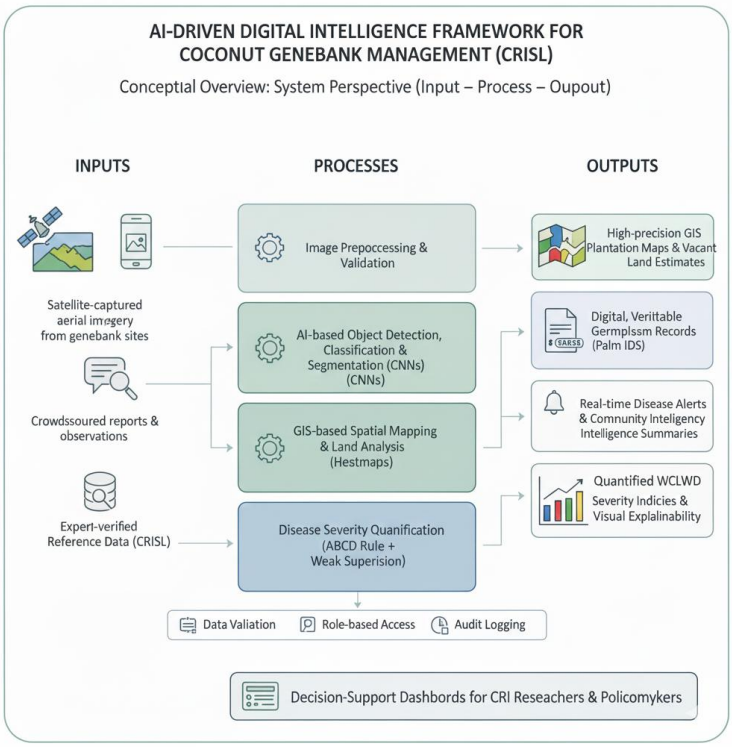
Processes

- Image preprocessing and quality validation
- AI-driven object detection, image classification, and segmentation
- Disease severity quantification using the ABCD rule and weak supervision
- Data validation, analytics, role-based access control, and audit logging

Outputs

- High-precision GIS plantation maps and vacant land estimates
- Verifiable digital coconut germplasm records
- Decision-support dashboards for CRI researchers and policymakers
- Quantified WCLWD severity indices and explainable visual maps

The conceptual diagram illustrates data flow from data acquisition sources such as drones, mobile devices, and community inputs to AI processing layers, including image processing, machine learning models, and GIS analysis. The processed information is stored and visualized through secure data management layers to provide decision-support services to CRISL authorities. Clear separation of input, processing, and output layers ensures modularity, scalability, and future integration with soil, climate, and yield intelligence systems.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The Coconut Research Institute (CRI) and coconut gene bank facilities function as the project's primary knowledge and resource centers. CRI technical documents, research publications, and interactions with Coconut Development Officers (CDOs) and field officers provide technical guidance on coconut diseases (including WCLWD), disease severity indicators, symptom stages, coconut varieties, and gene bank conservation practices. This insight ensures compliance to recognized agricultural standards while supporting all system components, such as disease analysis, gene bank monitoring, community data interpretation, and decision-support reporting. The project team's academic background and practical expertise in online systems, data analytics, and AI-based solutions allow the technological execution. Standard web frameworks are used to develop community platforms and backend services, but analytics and data processing.

The project team's academic background and practical knowledge of online systems, data analytics, and artificial intelligence (AI) technologies enable the technological execution. Standard web frameworks are used for developing community platforms and backend services, and Pandas and NumPy are used to implement data processing and analytics in Python. Convolutional neural networks and image embedding models are used in image-based analysis, and one-shot and few-shot learning strategies allow for effective operation with little labelled data. Structured parameters are extracted from reports and conversations using lightweight NLP techniques like Named Entity Recognition (NER) and embedding-based text analysis. Docker and cloud-based storage are used to manage system deployment, while Chart.js and Power BI are used to create visualization and analytics dashboards.

Data availability is ensured through a combination of community-generated and institutional sources. A mix of institutional and community-generated sources ensures data availability. The community platform is utilized to accumulate field-level data in the form of images, structured reports, discussions, and survey answers. CRI and gene bank documentation are the foundations of institutional databases pertaining to disease characteristics, coconut varieties, and gene bank records. While low-data learning techniques reduce the dependence on huge annotated datasets, automated prompts and rule-guided parameter extraction aid in standardization incoming data. Scalable collection of information is supported by this method for both gene bank condition monitoring and disease surveillance.

The project handles non-sensitive agricultural data related to field observations and plant conditions from an ethical perspective. Data utilization is restricted to research, monitoring, and decision support, and user engagement is entirely optional. Standard data protection and access control procedures are used, and institutional data access complies to authorized guidelines. Formal ethical clearance is not necessary because the project does not involve intrusive, medical, or personal data, and responsible data governance procedures are followed throughout the research.

9. Objectives and Novelty

Main Objective **To develop an AI-enabled integrated intelligence framework for coconut genebank management and crowdsourced surveillance, enabling automated WCLWD severity assessment to support data-driven decision-making in Sri Lanka.**			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22904164	GIS-Driven Land Utilization Analysis for Coconut Gene banks	A GIS-centric land analysis system integrates satellite-captured/drone captured aerial imagery with geospatial processing to create accurate plantation maps of CRISL gene banks. Detected coconut trees are geo referenced and mapped within GIS layers to analyze spatial distribution, tree density, and land utilization. The system generates GIS-based heatmaps to identify high density, low-density, and vacant areas, calculates exact plantable land parcels, and predicts optimal locations and capacity for new coconut plantations.	• First GIS centric, satellite/drone-based land utilization analysis for coconut gene banks in Sri Lanka • Tight integration of geo referenced satellite/drone imagery with GIS layers for accurate plantation mapping • Automated spatial analytics with heatmaps and exact area calculations, replacing manual surveys
IT22236500	A Secure Image Classification System for Coconut	Develop a centralized web-based image classification system for the Coconut Research Institute (CRI) of Sri Lanka to enable digital identification of coconut	• First CRI-aligned AI framework in Sri Lanka for coconut germplasm

	Germplasm Identification	germplasm across regional gene banks. Authorized CRI personnel upload palm and nut images, which are assigned unique Palm IDs and automatically analyzed using CNN based models. Extracted visual parameters, AI predictions, confidence scores, and image metadata are then automatically stored in a centralized digital gene bank database for secure and standardized record management. • CRI Aligned Morphological Parameter Extraction • Role-Based Access Control and Audit Management	identification using palm and nut images. • Automatic extraction of CRI-defined morphological parameters from images instead of manual visual assessment. • Application of few-shot learning to address limited labeled germplasm image availability at CRI. • Secure, role-based, and traceable digital gene-bank record generation from AI-processed field images. • Use of explainable AI (Grad-CAM) to improve expert trust in AI-based germplasm classification.
IT22549204	Crowdsourcing-Based Community Intelligence Platform for Coconut Disease and Conservation Surveillance	A crowdsourcing-based community intelligence platform that collects real-time field data (images, reports, discussions) from farmers, field officers, researchers, and students, and transforms it into structured, confidence scored agricultural intelligence. The system applies low data AI techniques (one shot and few-shot learning) to analyze coconut disease conditions (including WCLWD)	• Integrates crowdsourced community data for both coconut disease surveillance and coconut gene bank monitoring within a single platform. • Applies one shot and few shot learning to enable AI analysis under limited labelled data conditions. • Transforms unstructured posts, images, and discussions into

		and coconut gene bank data, constructs knowledge mappings beyond simple location-based mapping, and provides automated summaries and visual insights to support timely decision-making by CRI and related authorities.	structured agricultural intelligence. • Introduces a conceptual knowledge mapping approach beyond traditional location-based mapping. • Provides Sri Lanka–specific insights to support CRI decision making without replacing expert evaluation.
IT22088314	Weligama Coconut Leaf Wilt Disease (WCLWD) Severity Assessment using ABCD Rule	An automated disease detection and severity quantification framework for Weligama Coconut Leaf Wilt Disease (WCLWD) using Weakly Supervised Semantic Segmentation (WSSS) combined with the ABCD severity assessment rule. The framework eliminates the need for manual pixel-level annotation, accurately identifies diseased regions, and quantitatively measures disease severity under real-world field conditions.	• First application of automatic pixel level for WCLWD using weak supervision. • Elimination of expert-dependent manual labeling. • Integration of the ABCD rule for coconut disease severity assessment. • Improved differentiation between WCLWD and nutrient deficiencies.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

The topic is recommended and the 4 functions are consisting research areas to develop.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Sanvitha	Kartburlarachchi	
Co-Supervisor	Ms.	Karthuja	Rajendran	
External Supervisor	Mr.	Oshan	Thilakarathne	Oshan

Summary of external supervisor's (if any) experience and expertise

Research Officer, Genetics and plant breeding division, Coconut Research Institute, Lunuwila, Sri Lanka